

CU
A

Senior Project Design

Course No.: 122-495-01

Instructor: J. Roberts

*Database Management In Internet Oriented
Environment*

By

SoftSystems - Igor Smirnov

Spring, 2000.

MILESTONE EVALUATION SHEET

MILESTONE NO.

1

DATE: 02/21/00	COURSE NO.: 122-495-01
	INSTRUCTOR: J. Roberts

TEAM NO. OR NAME	TEAM MEMBERS
SoftSystems	
	Igor Smirnov

Student Comments:

Instructor Comments:

Score:



SoftSystems

Igor Smirnov

Phone: 863-7633 Fax: 838-1115

REQUEST FOR INFORMATION SYSTEM SERVICES

DATE OF REQUEST	SERVICE REQUESTED FOR DEPARTMENT(S)
02/06/2000	Computer Science Department, Charleston Southern University(CSU)

SUBMITTED BY (key user contact)	EXECUTIVE SPONSOR (funding authority)
Name Dr. Jim roberts	Name Dr. Jim roberts
Title Professor	Title Professor
Office Professor	Office Professor
Phone 7004	Phone 7004

TYPE OF SERVICE REQUESTED:

- | | |
|---|---|
| ☒ Information Strategy Planning | ☐ Existing Application Enhancement |
| ☐ Business Process Analysis and Redesign | ☐ Existing Application Maintenance (problem fix) |
| ☒ New Application Development | ☐ Not Sure |
| ☒ Other (please specify): <u>Design of a web page</u> | |

BRIEF STATEMENT OF PROBLEM, OPPORTUNITY, OR DIRECTIVE

The Computer Science Department of Charleston Southern University would like to research and explore the capabilities of database management through an Internet oriented environment. In order to do so, the Department is calling for a design and implementation of an Internet information system. Using new technologies currently available, it will demonstrate the use of programming languages and tools in developing an Internet oriented database system.

The System would be demonstrated on an independent Internet server. If satisfied the expectations of the Department, it would be used as a reference to future Internet oriented classes and projects.

BRIEF STATEMENT OF EXPECTED SOLUTION

The development of a web page or series of web pages focused on database technologies. The Page(s) will be served from an independent Internet server provider. They will demonstrate the way databases can be accessed and manipulated through Internet. The website will be attractive looking and easy-to-follow.

ACTION (ISS Office Use Only)

- | | |
|--|---------------------------------|
| ☐ Feasibility assessment approved | Assigned to _____ |
| ☐ Feasibility assessment waived | Approved Budget \$ _____ |
| | Start Date _____ Deadline _____ |
| ☐ Request delayed | Backlogged until date: _____ |
| ☐ Request rejected | Reason: _____ |

Authorized Signatures:

Chair, ISS Executive Steering Body

Project Executive Sponsor

PROBLEM STATEMENTS

PROJECT: Design of a Internet oriented database	PROJECT MANAGER: Igor Smirnov
CREATED BY: SiftSystems – Igor Smirnov	LAST UPDATED BY: Igor Smirnov
DATE CREATED: Tuesday, February 15, 2000	DATE LAST UPDATED: Thursday, April 06, 2000

Brief Statements of Problem, Opportunity, or Directive	Urgency	Visibility	Annual Benefits	Priority or Rank	Proposed Solution
1. No design in place for a web page	3 months	High	Unknown	1	New Development
2. No design in place for a database	8 months	High	Unknown	2	New Development
3. No web page server in place	5 months	Low	Unknown	4	Free internet server
4. Opportunity to build an Internet oriented database	8 months	High	Unknown	3	New Development
5. Opportunity to deploy and use new technologies for later use of the Department	7 months	High	Unknown	5	Future, updated version of the web page

MILESTONE EVALUATION SHEET

MILESTONE NO.

2

DATE: 02/28/00	COURSE NO.: 122-495-01
	INSTRUCTOR: J. Roberts

TEAM NO. OR NAME	TEAM MEMBERS
SoftSystems	
	Igor Smirnov

Student Comments:

Instructor Comments:

Score:

PROBLEMS, OPPORTUNITIES, OBJECTIVES AND CONSTRAINTS MATRIX

Project: Design of a Internet oriented database	Project Manager: Igor Smirnov
Created by: SoftSystems – Igor Smirnov	Last Updated by: Igor Smirnov
Date Created: Tuesday, February 15, 2000	Date Last Updated: Thursday, April 06, 2000

CAUSE AND EFFECT ANALYSIS		SYSTEM IMPROVEMENT OBJECTIVES	
Problem or Opportunity	Causes and Effects	System Objective	System Constraint
1. Need to decide what technologies will be used as a base for the project. 2. Finding a fast and free independent web server	1. Currently, many technologies can be used in order to put the database in place with many pros and Cons of each. 2. Most of free Internet servers are not as fast as the commercial ones.	1. Find a technologies comfortable and advanced enough to offer the best performance for the project 2. Find a web server with a fast and reliable internet connection to fulfill the needs of the project	1. No known constraints.

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3

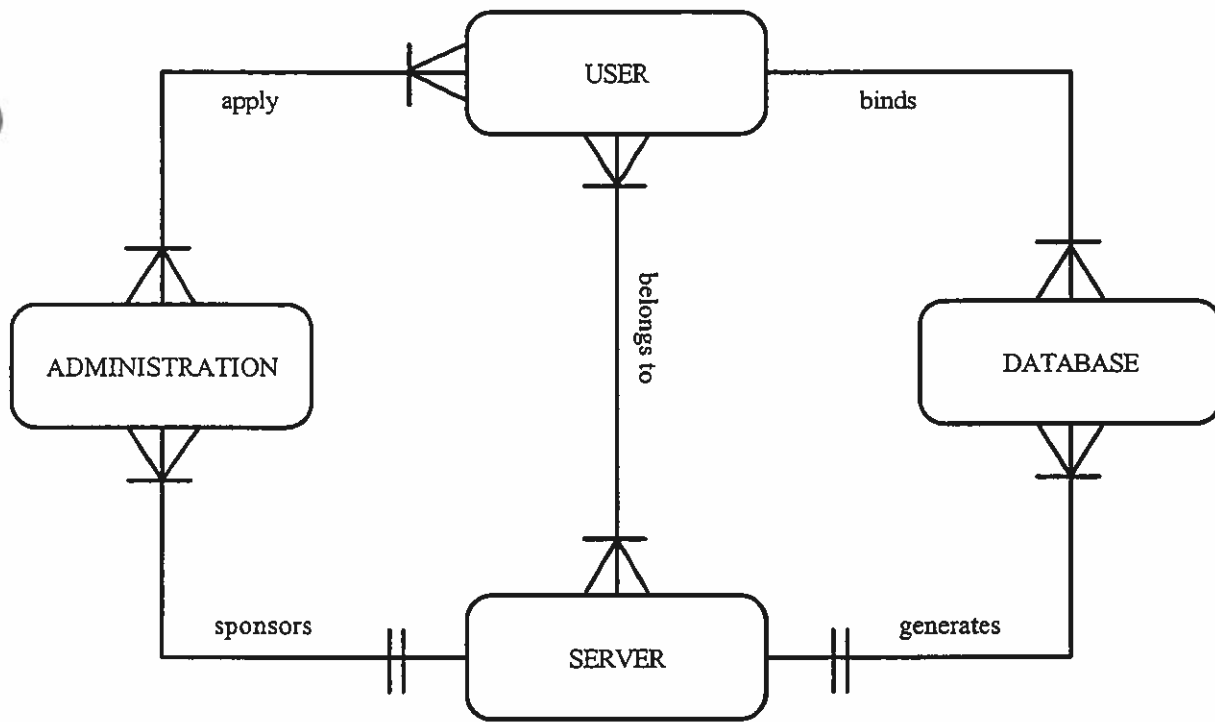
DATE: 03/06/00	COURSE NO.: 122-495-01
	INSTRUCTOR: J. Roberts

TEAM NO. OR NAME	TEAM MEMBERS
SoftSystems	
	Igor Smirnov

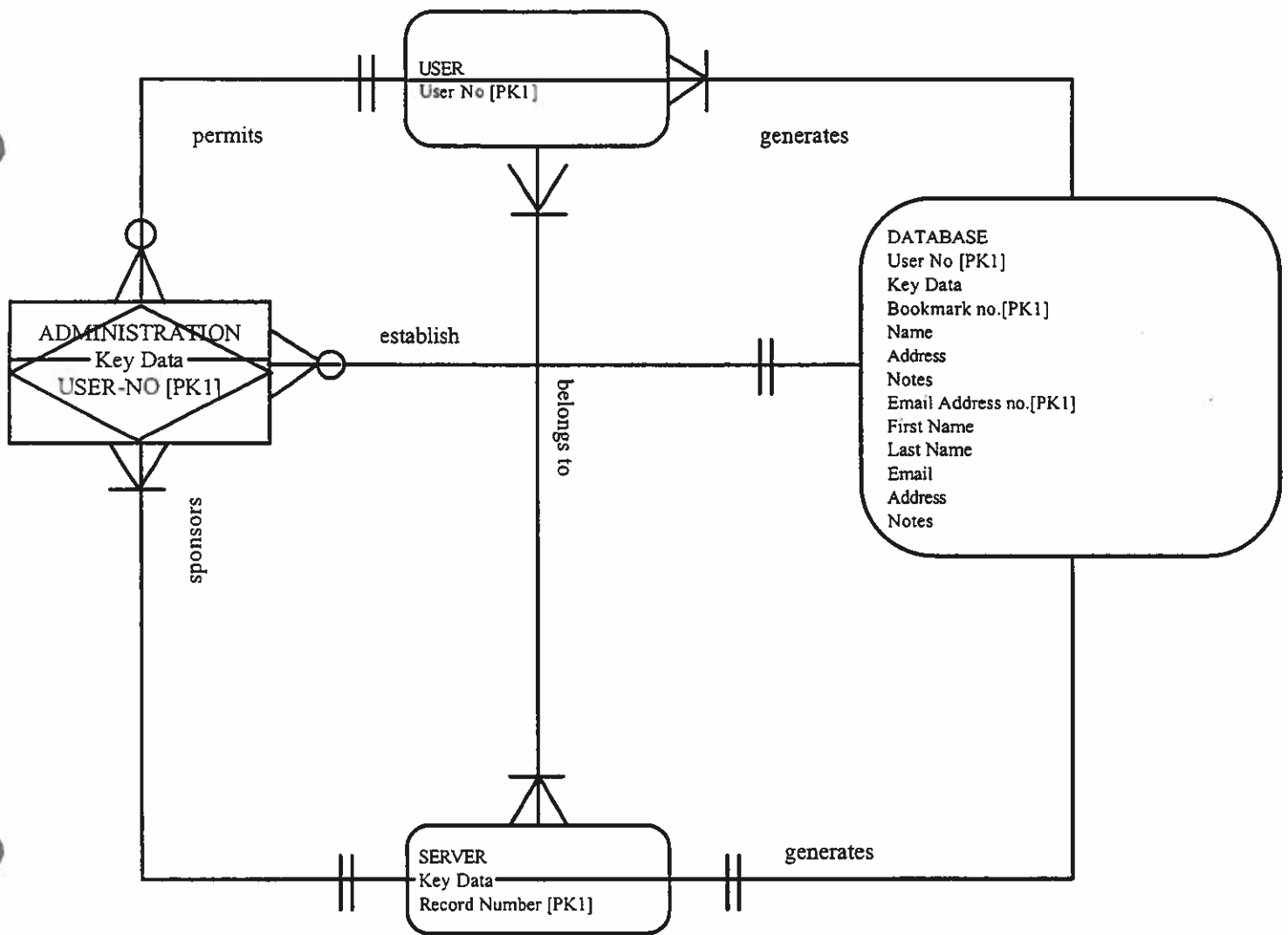
Student Comments:

Instructor Comments:

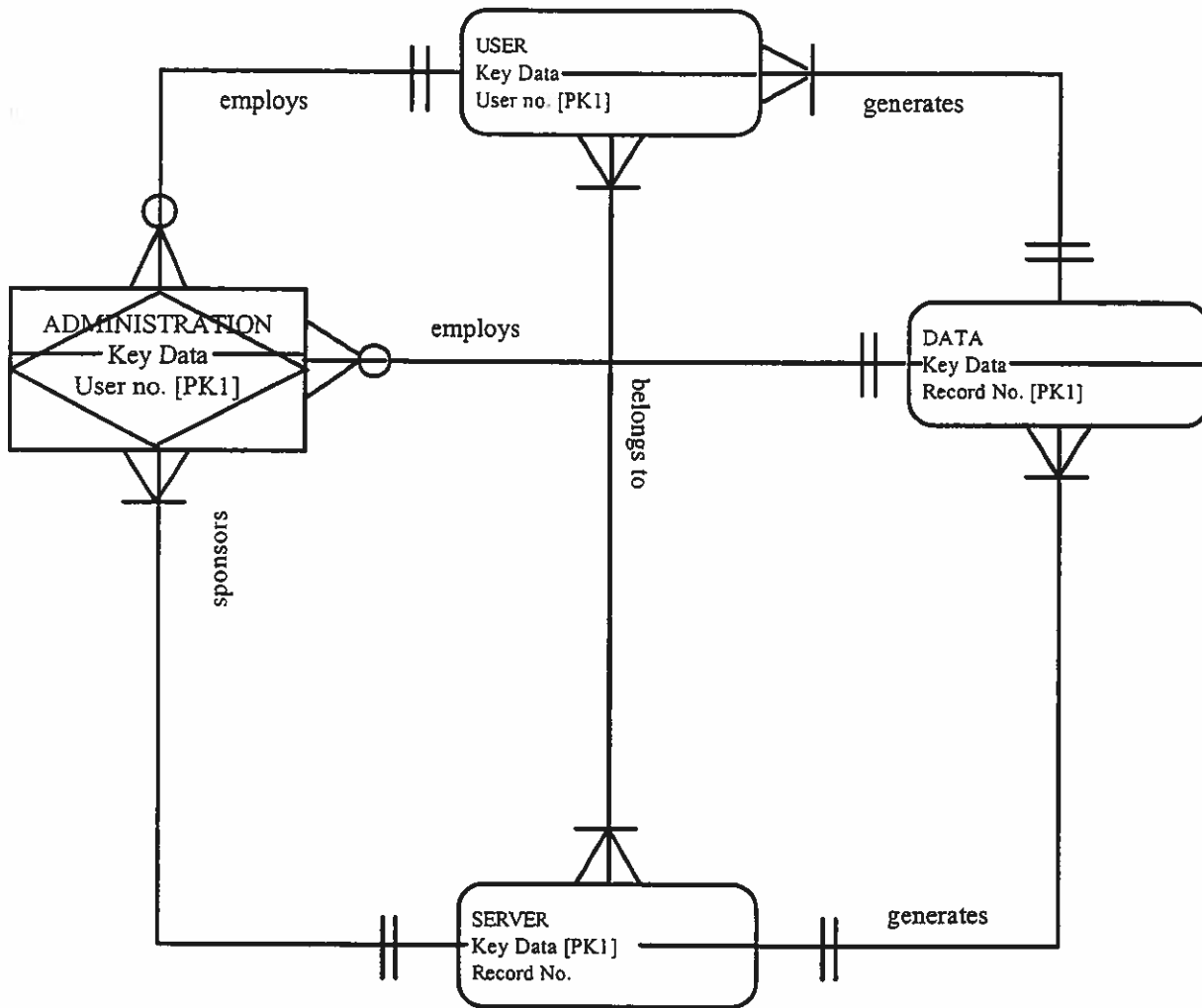
Score:



SoftSystems
Entity Relationship Context Diagram
Igor Smirnov
March 9, 2000.



SoftSystems
Fully Attributed Data Model
Igor Smirnov
March 9, 2000.



SoftSystems
Key Based Data Model
Igor Smirnov
March 10, 2000.

Entity/Definition Matrix

ENTITY	BUSINESS DEFINITION
Users	Users are the ones who put their records online
Server	Is the database processing system
Administration	The database and server management crew
Database	Database of users records

MILESTONE EVALUATION SHEET

MILESTONE NO.

4

DATE: 03/13/00	COURSE NO.: 122-495-01
	INSTRUCTOR: J. Roberts

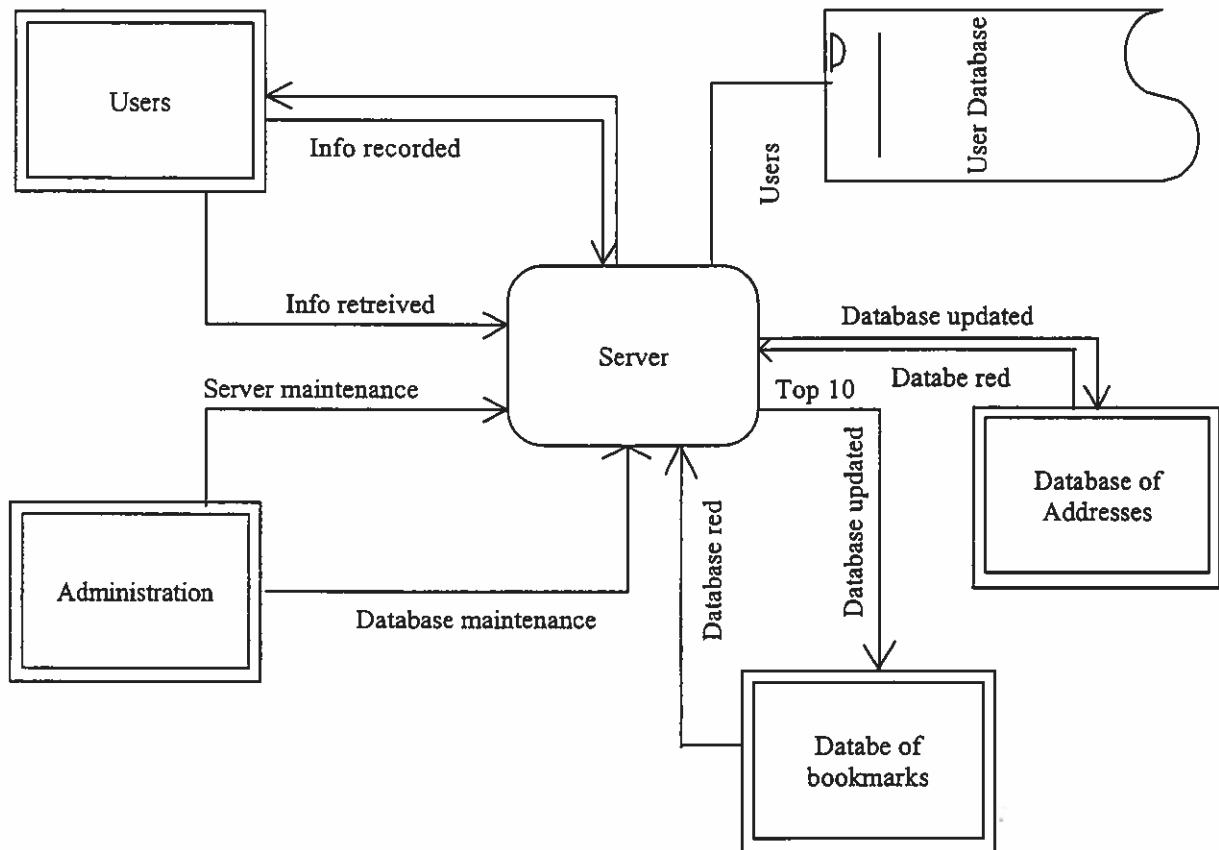
TEAM NO. OR NAME	TEAM MEMBERS
SoftSystems	
	Igor Smirnov

Student Comments:

Instructor Comments:

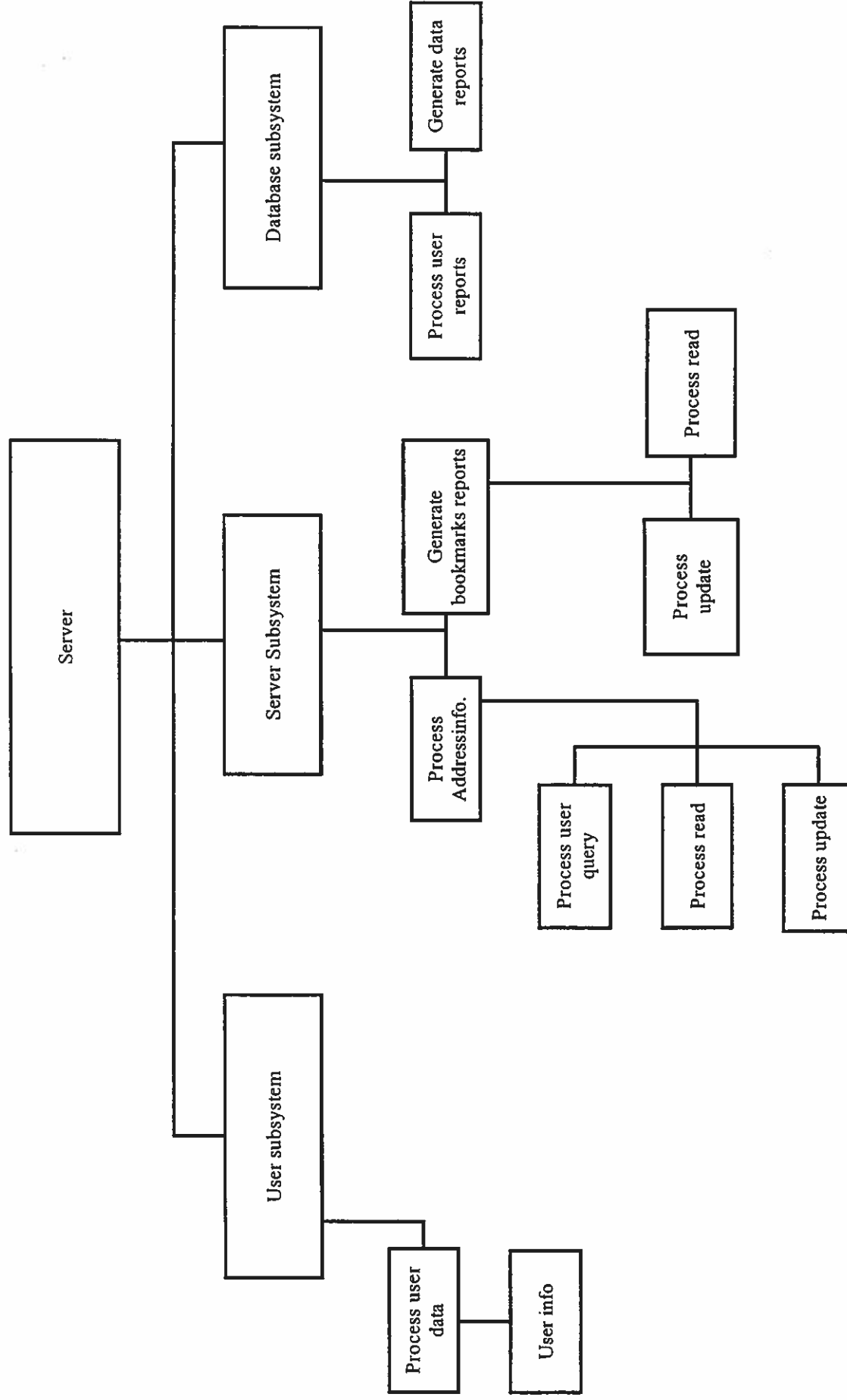
Score:

CONTEXT DIAGRAM



SoftSystems
Context Diagram
Igor Smirnov
March 12, 2000.

EVENT DECOMPOSITION DIAGRAM



EVENT LIST

Event Description	Trigger (Inputs)	Respons (Outputs)
Address lookup	User request a list of addresses	Addresses table report
Bookmark lookup	User request a list of bookmarks	Bookmark table report
Add bookmark/address	Add request	Database updated
Remove bookmark/address	Remove request	Database updated

MILESTONE EVALUATION SHEET

MILESTONE NO.

5

DATE: 03/20/00	COURSE NO.: 122-495-01
	INSTRUCTOR: J. Roberts

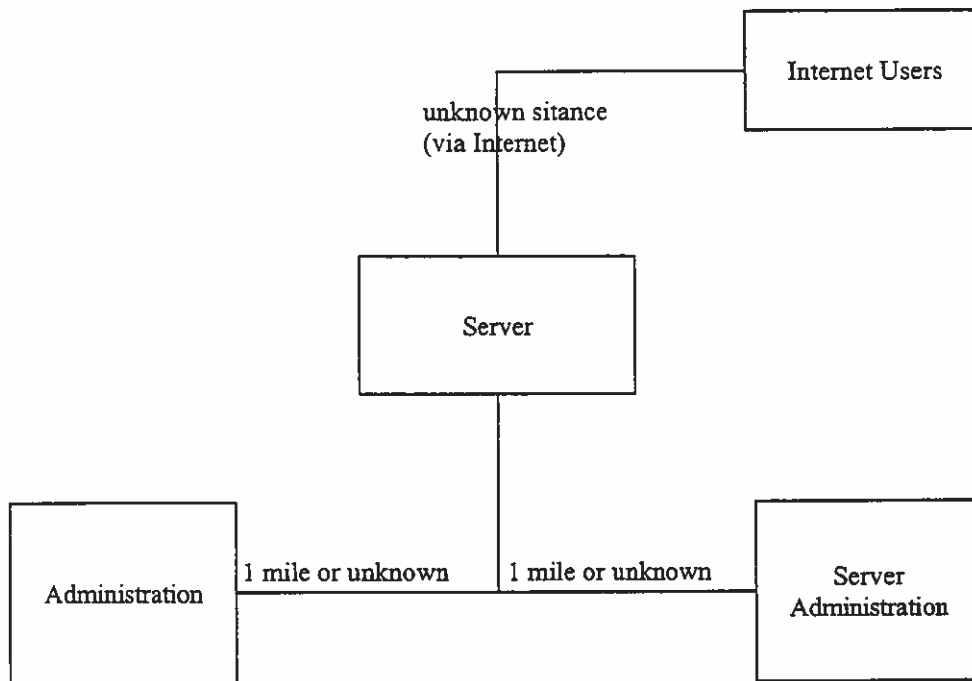
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SoftSystems	
	Igor Smirnov

Student Comments:

Instructor Comments:

Score:

LOCATION CONECTIVITY DIAGRAM



SoftSystems
LCD
March 20, 1999.

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DATE: 03/28/00	COURSE NO.: 122-495-01
	INSTRUCTOR: J. Roberts

TEAM NO. OR NAME	TEAM MEMBERS
SoftSystems	
	Igor Smirnov

Student Comments:

Instructor Comments:

Score:

Characteristics	Candidate 1	Candidate 2
Portion of System Computerized Brief description of that portion of the system that would be computerized in this candidate.	Internet Information System. Microsoft Access.	Same as candidate 1.
Benefits Brief description of the business benefits that would be realized for this candidate.	Provides easy-to-use HTML forms for configuring and administering the server. Ease of database management tools use.	Fully supports user required processes for the demonstration to the COINS department.
Servers and Workstations A description of the servers and workstations needed to support this candidate.	Fast Windows NT or Linux class servers.	Same as Candidate 1.
Software Tools Needed Software tools needed to design and build the candidate (e. g., database management system, emulators, operating systems, languages, etc.). Not generally applicable if applications software packages are to be purchased.	Microsoft Office 2000 for customization of package to provide report writing and integration.	MS Visual Basic 6.0 ASP Internet Explorer
Application Software A description of the software to be purchased, built, accessed, or some combination of these techniques.	Package Solution.	Custom Solution.
Method of Data Processing Generally some combination of: on-line, batch, deferred batch, remote batch, and real-time.	Client/Server	Same as Candidate 1.
Output Devices and Implications A description of output devices that would be used, special output requirements, (e.g. network, preprinted forms, etc.), and output considerations (e.g., timing constraints).	An interactive web page that will allow the transfer of documents requested from a browser with HTTP. For configuration and administration, a Web browser such as Microsoft Internet Explorer or Netscape Navigator.	Same as Candidate 1.
Input Devices and Implications A description of Input methods to be used, input devices (e.g., keyboard, mouse, etc.), special input requirements, (e.g. new or revised forms from which data would be input), and input considerations (e.g., timing of actual inputs).	Keyboard and Mouse	Same as Candidate 1.
Storage Devices and Implications Brief description of what data would be stored, what data would be accessed from existing stores, what storage media would be used, how much storage capacity would be needed, and how data would be organized.	SQL Server DBMS	Virtual internet server.

Feasibility Criteria	Weight	Candidate 1	Candidate 2
Operational Feasibility Functionality. A description of to what degree the candidate would benefit the organization and how well the system would work. Political. A description of how well received this solution would be from both user management, user, and organization perspective.	30%	Internet Information Server. It fully supports user-required functionality. Score: 100	Microsoft SQL server business processes might need modifying to take advantage of software functionality. Score: 80
Technical Feasibility Technology. An assessment of the maturity, availability (or ability to acquire), and desirability of the computer technology needed to support this candidate. Expertise. An assessment to the technical expertise needed to develop, operate, and maintain the candidate system.	30%	Microsoft Office 2000 has been around for a while now and should be the best performance/stability option to take. The cost of a package is affordable. Score: 100	MS Visual Basic is proven a very good environment for database management and business programs implementation. Score: 100
Economic Feasibility Cost to develop: Payback period (discounted): Net present value: Detailed calculations:	30%	\$0. \$0. Undefined (\$) Not applicable Score: 100	Score: 95
Schedule Feasibility An assessment of how long the solution will take to design and implement.	10%	Less than six months. Score: 90	Less than six months Score: 90
Ranking:	100%	90	90

MILESTONE EVALUATION SHEET

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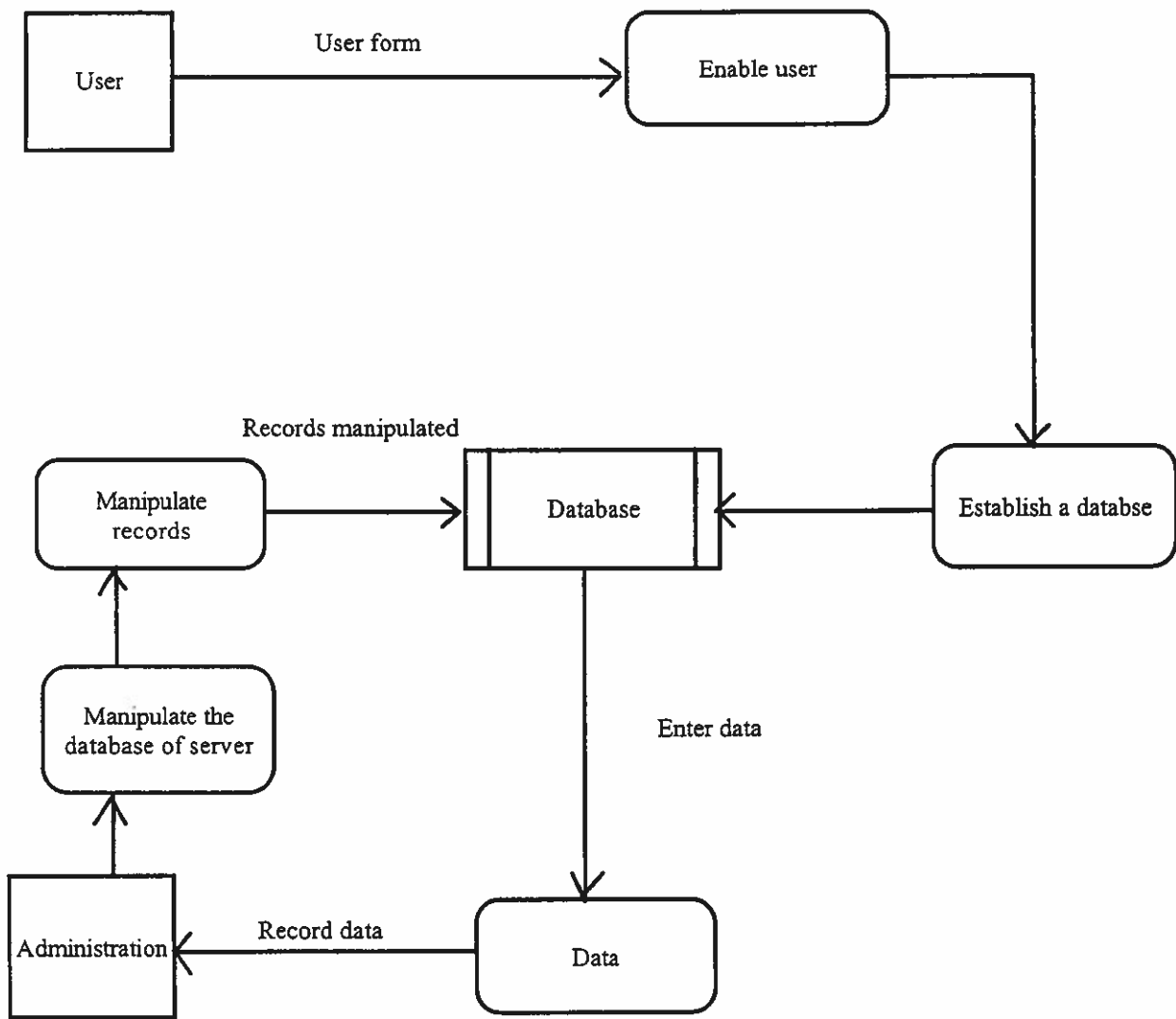
DATE: 04/04/00	COURSE NO.: 122-495-01
	INSTRUCTOR: J. Roberts

TEAM NO. OR NAME	TEAM MEMBERS
SoftSystems	
	Igor Smirnov

Student Comments:

Instructor Comments:

Score:



SoftSystems
Physical Data Flow Diagram
March 14, 2000.

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DATE: 04/11/00	COURSE NO.: 122-495-01
	INSTRUCTOR: J. Roberts

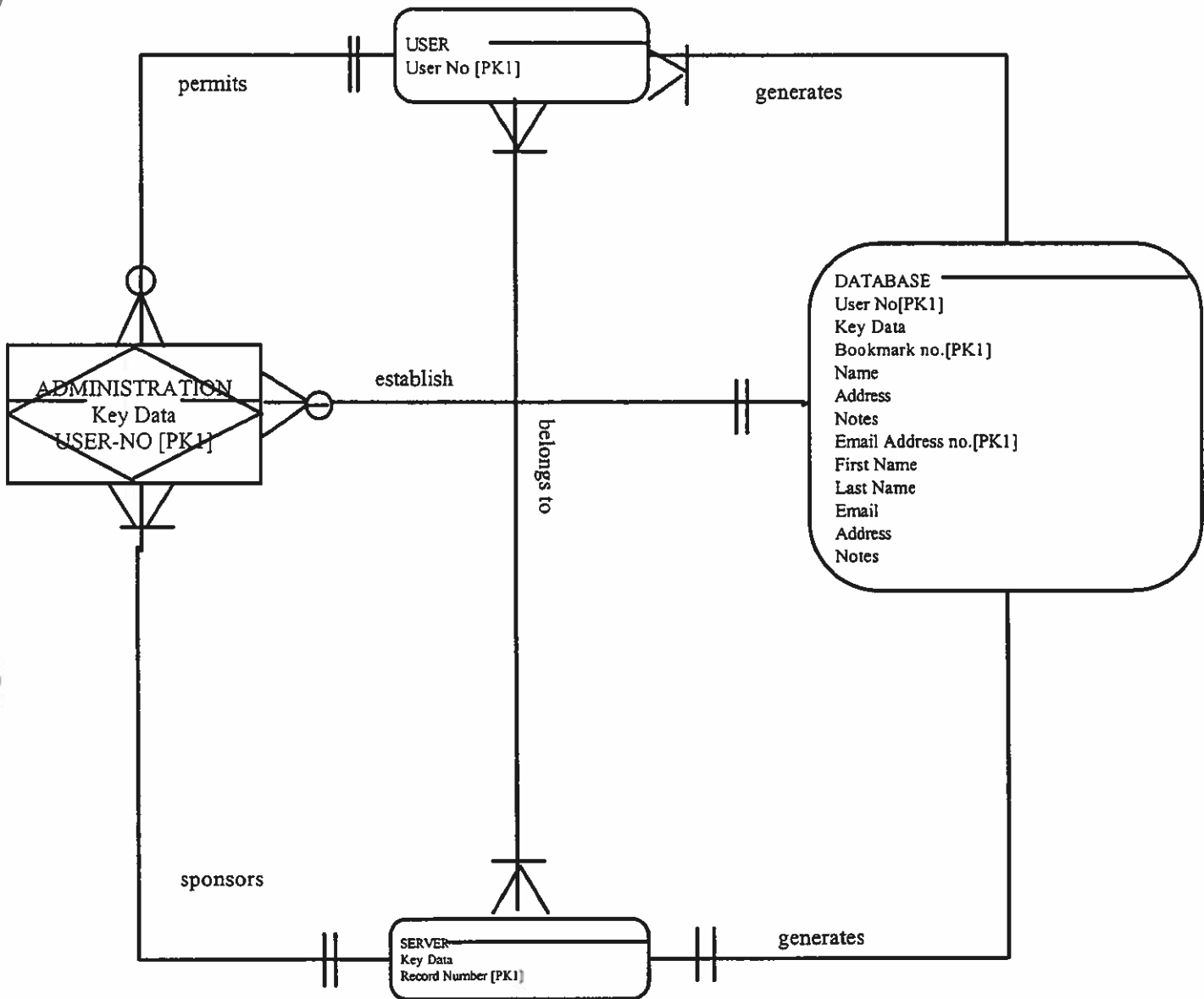
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SoftSystems	
	Igor Smirnov

Student Comments:

Instructor Comments:

Score:

THIRD NORMAL FORM



SoftSystems
Normalized Relations
Igor Smirnov
March 27, 2000.

COMPLETE TABLE DEFINITIONS

User Table

User No. [PK1]

Name

Bookmarks Table

PK Bookmark no; numerical, 4 characters

Name

HTTP Address

Notes

Address Table

PK Address no; numerical 4 characters

Name

Last Name

First Name

Email address

Notes

SoftSystems
CTF
March 28, 2000.

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MILESTONE NO.

9

DATE: 04/18/00	COURSE NO.: 122-495-01
	INSTRUCTOR: J. Roberts

TEAM NO. OR NAME	TEAM MEMBERS
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	Igor Smirnov

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Instructor Comments:

Score:

Enter Bookmark details	
Name	CSU
Address	http://www.csuniv.edu
Notes	
Some notes	

Internal Sample form
April 1, 2000.
SoftSystems

Enter Email details	
Name	Igor
Email	sigor1@usa.net
Last Name	Smirnov
First Name	Igor
Notes	
Some notes	

Internal Sample form
April 1, 2000.
SoftSystems

MILESTONE EVALUATION SHEET

MILESTONE NO.

10

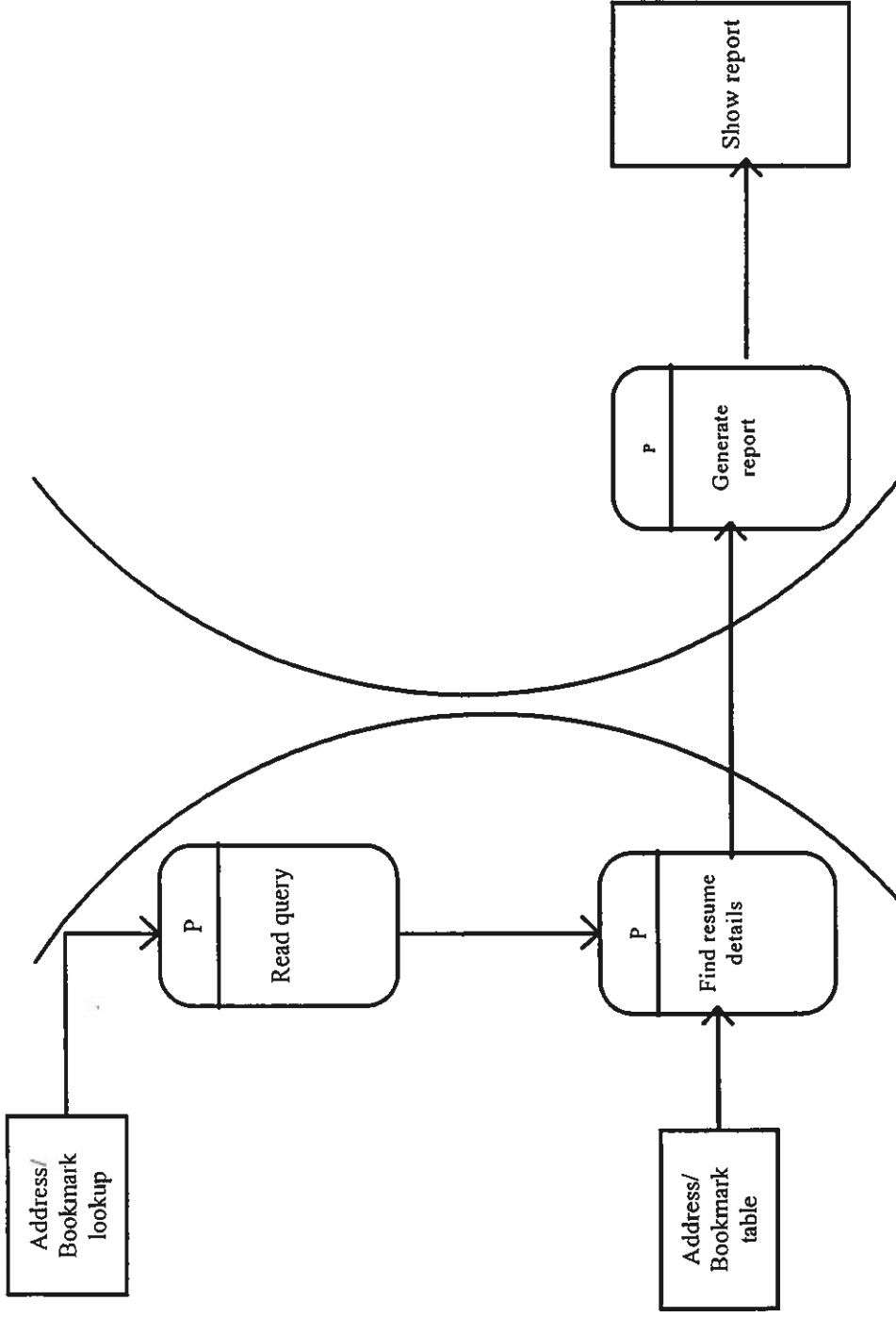
DATE: 04/25/00	COURSE NO.: 122-495-01
	INSTRUCTOR: J. Roberts

TEAM NO. OR NAME	TEAM MEMBERS
SoftSystems	
	Igor Smirnov

Student Comments:

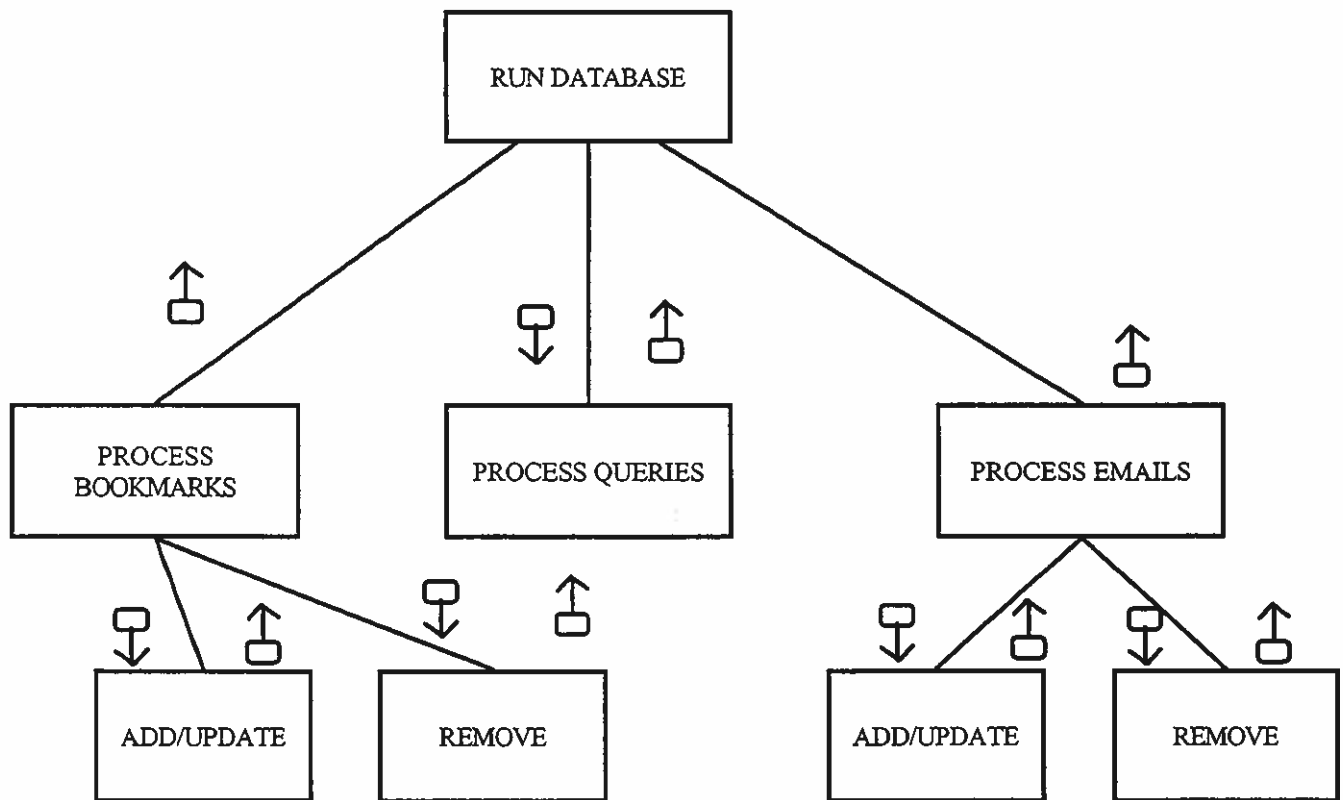
Instructor Comments:

Score:



Partitioned DFD
Systems Software
April 26, 2000.

(Last updated April 26, 2000.)



Structure Chart
Igor Smirnov
SoftSystems
April 26,2000

(Last updated April 29,2000)

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11

DATE: 04/22/00	COURSE NO.: 122-495-01
	INSTRUCTOR: J. Roberts

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SoftSystems	
	Igor Smirnov

Student Comments:

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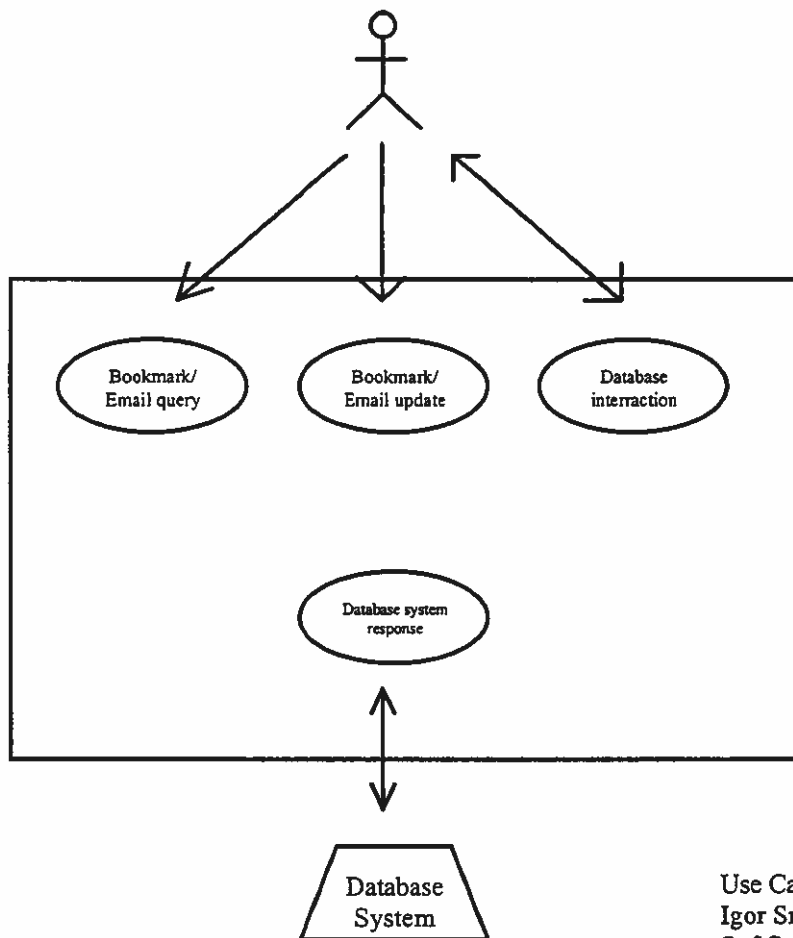
USE CASE

Author: SoftSystems

Date: 04/27/2000

USE CASE NAME:	Database Add/Remove/Update
ACTOR:	User
DESCRIPTION:	Describes the process when user calls for action on the database. He adds to, removes or updates the information contained in the tables.
NORMAL COURSE:	1. The user in interaction with the system chooses one of the actions available. 2. Page with the table responsible for recording the user's action is loaded. 3. User is able to enter all necessary information to record his action. 4. Action is recorded in the system and the database is updated. 5. The system is ready for further interaction with the user.
PRECONDITION:	User is allowed to use the system.
POST-CONDITION:	Information is recorded, database updated. In case of problems, system's administrator can resolve the problem.
ASSUMPTIONS:	None

Use Case Description
Igor Smirnov
SoftSystems
April 27,2000



Use Case Model Diagram
Igor Smirnov
SoftSystems
April 30,2000.

(Last updated April 30,2000.)

MILESTONE EVALUATION SHEET

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12

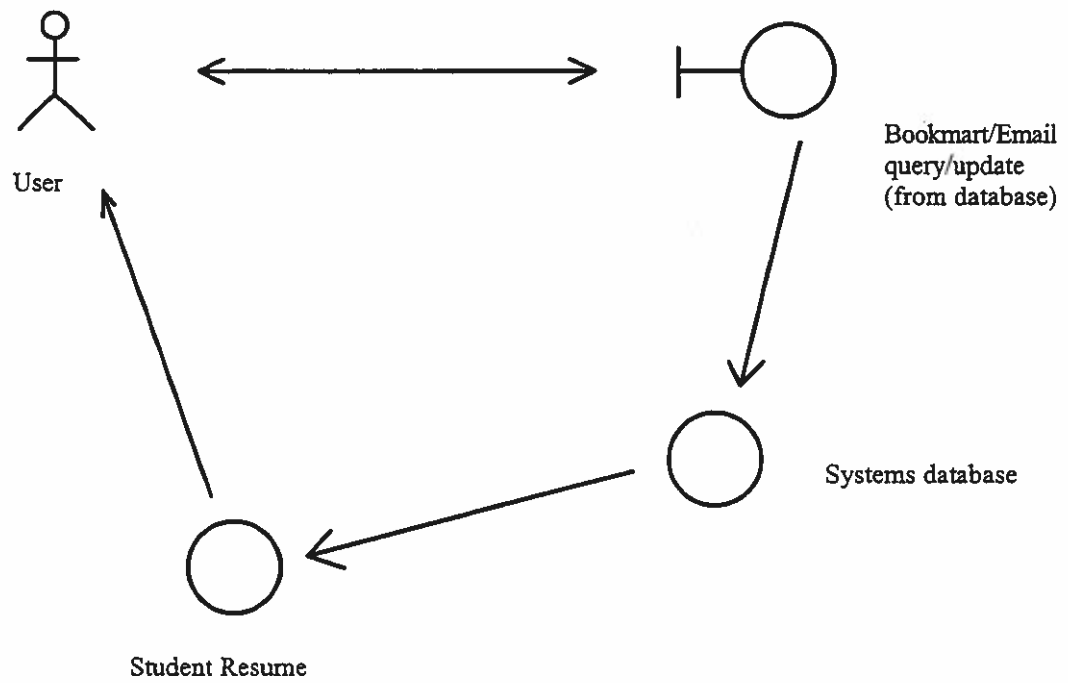
DATE: 04/29/00	COURSE NO.: 122-495-01
	INSTRUCTOR: J. Roberts

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SoftSystems	
	Igor Smirnov

Student Comments:

Instructor Comments:

Score:



Ideal Object Model
Igor Smirnov
SoftSystems
April 30,2000.

(Last updated April 30,2000.)